

END-TO-END DATA ANALYTICS PROJECT

Customer360 Analytics Platform

Medallion Architecture • Databricks • Power BI

Databricks

SQL

PySpark

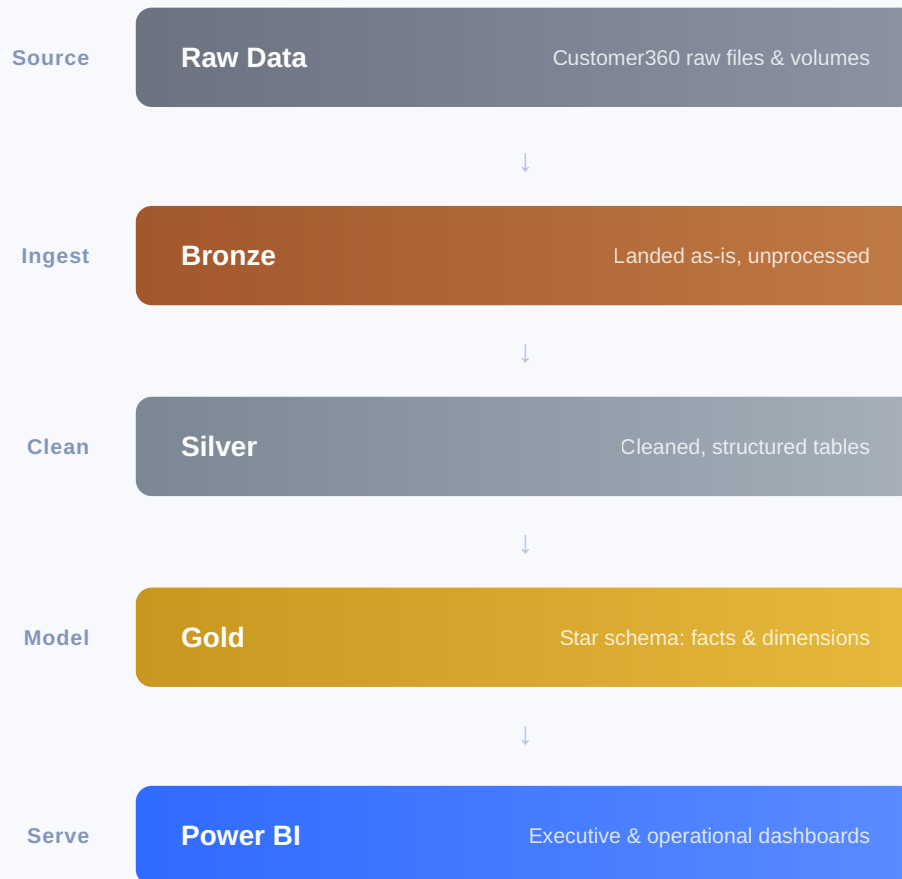
Power BI

DAX

Star Schema

Medallion Architecture

Medallion Architecture Pipeline



Pipeline Overview

Raw customer, order, product, clickstream and support-ticket data is ingested into Databricks and progressively refined through Bronze → Silver → Gold layers using PySpark and SQL, culminating in a governed Gold-layer star schema consumed directly by Power BI.

Tech Stack

Databricks

SQL

PySpark

Power BI

DAX

Star Schema

Unity Catalog

WORKSPACE & CATALOG EXPLORER

The screenshot shows the Databricks Workspace & Catalog Explorer interface. At the top, there's a 'workspace' header with a share button. Below it, tabs for 'Overview', 'Details', 'Permissions', 'Policies', and 'Workspaces' are visible. The 'Overview' tab is active, showing a 'Description' section with an 'AI generate' button and an 'Add' button. Below this is a search bar labeled 'Filter schemas' with a magnifying glass icon, and a result count of '4 schemas'. A table lists the schemas:

Name	Owner	Created at
customer360_gold	abdallaht12012005@gmail.com	Jul 23, 2026, 05:44 PM
customer360_silver	abdallaht12012005@gmail.com	Jul 22, 2026, 04:08 PM
default	_workspace_admins_workspace_747465...	Jul 21, 2026, 04:12 PM

UNITY CATALOG — GOLD / SILVER SCHEMAS

The screenshot shows the Databricks Unity Catalog — Gold / Silver Schemas interface. It displays a tree view of schemas and their tables:

- customer360_gold
 - Tables (7)
 - dim_customer
 - dim_date
 - dim_product
 - dim_support_agent
 - fact_clickstream
 - fact_orders
 - fact_support_tickets
- customer360_silver
 - Tables (5)
 - clickstream
 - customers
 - orders
 - products
 - support_tickets

SQL NOTEBOOK — GOLD LAYER TRANSFORMATION

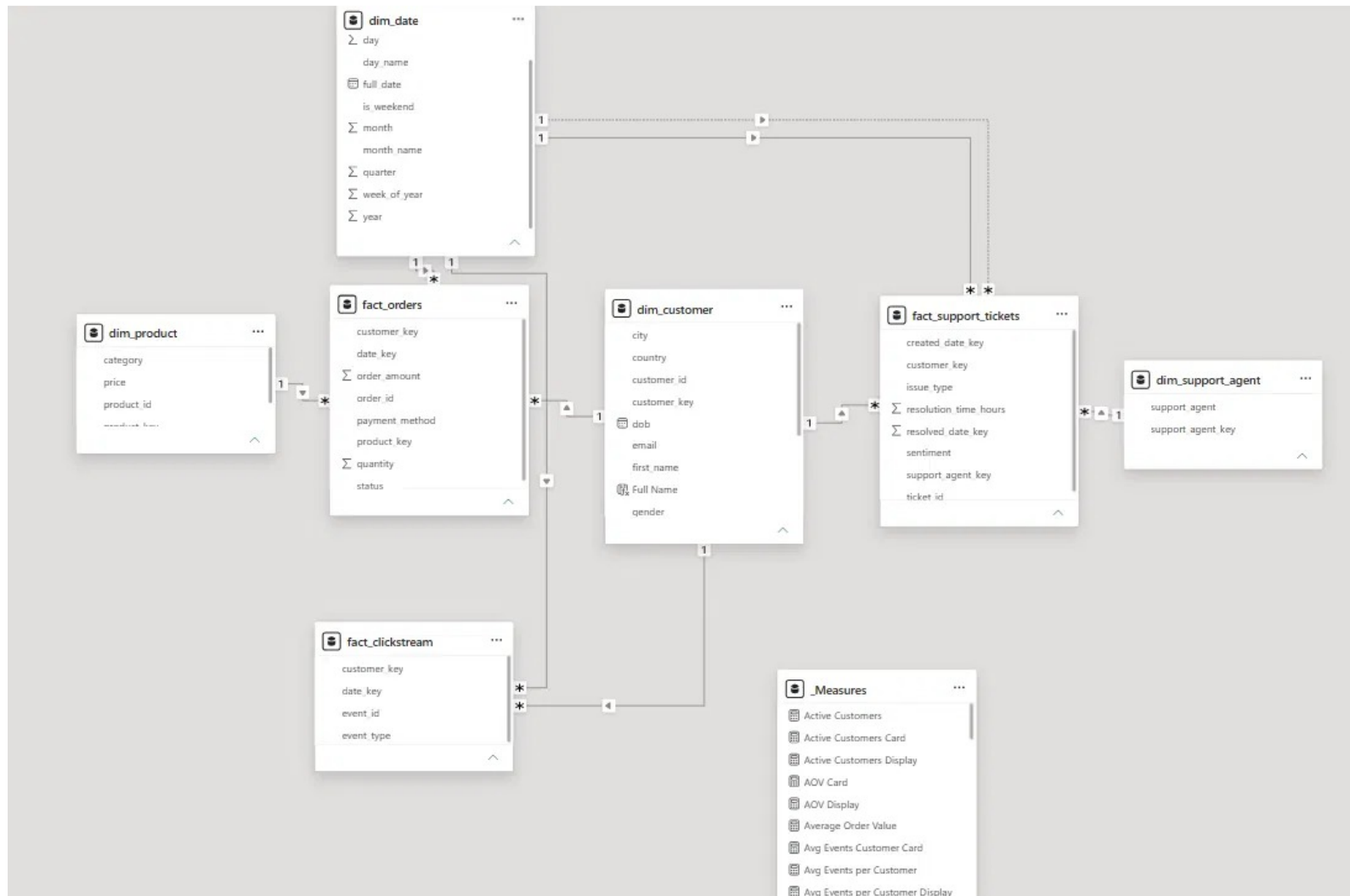
The screenshot shows a Databricks SQL Notebook with a transformation query. The query is as follows:

```
d AS full_date,
YEAR(d) AS year,
QUARTER(d) AS quarter,
MONTH(d) AS month,
DATE_FORMAT(d, 'MMMM') AS month_name,
DAY(d) AS day,
DATE_FORMAT(d, 'EEEE') AS day_name,
WEEKOFYEAR(d) AS week_of_year,

CASE
  WHEN DAYOFWEEK(d) IN (1,7) THEN 'Yes'
  ELSE 'No'
END AS is_weekend

FROM (
  SELECT EXPLODE(
    SEQUENCE(
      TO_DATE('2013-01-01'),
      TO_DATE('2027-12-31'),
```

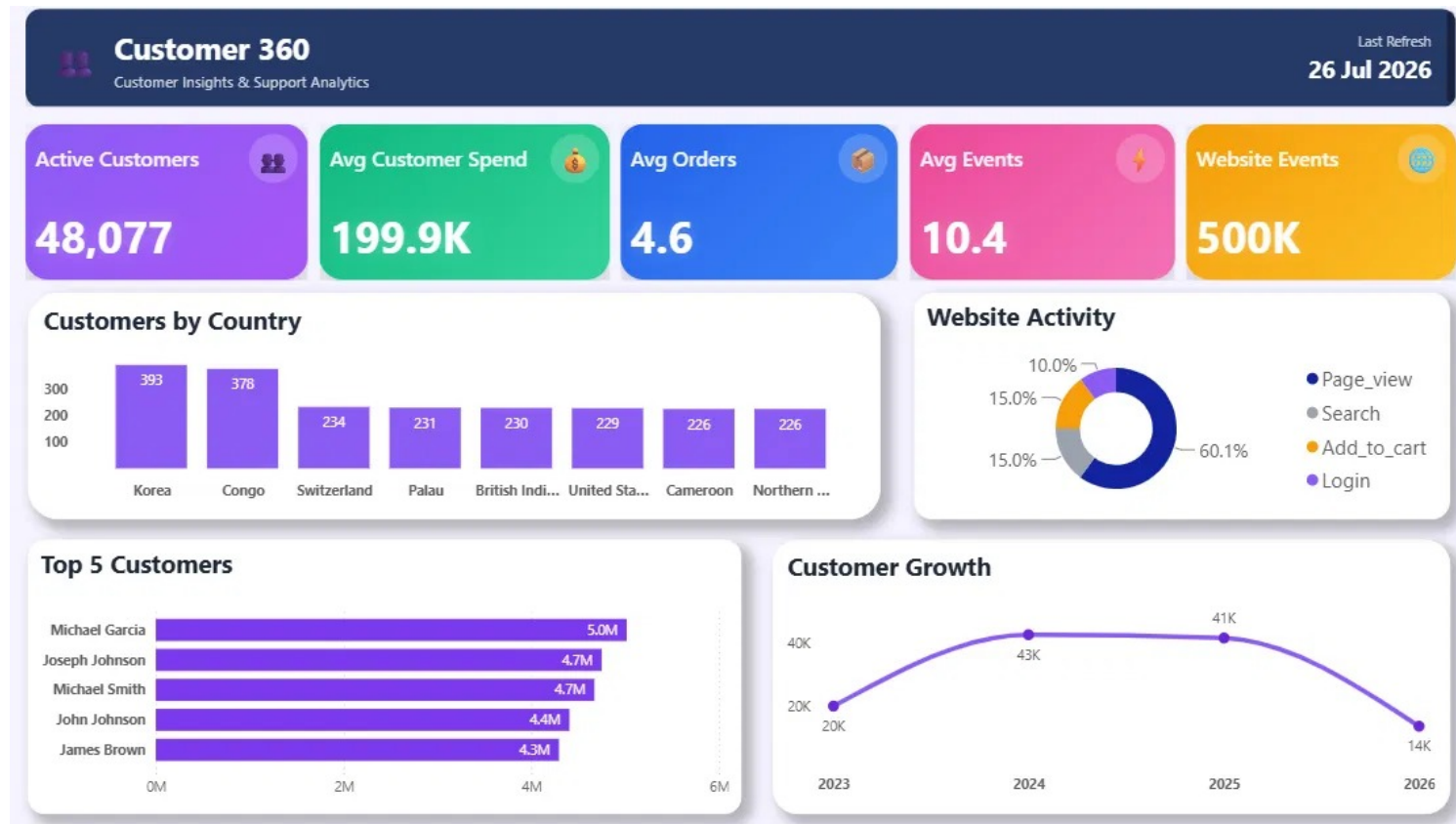
Power BI Data Model — Star Schema



Consolidated KPIs — 9.61B total revenue, 48.2K customers, 221.3K orders — refreshed daily for leadership review.

Revenue trended across 2023–2026 and segmented by product category, payment method and top-selling products.

Year-based slicers let stakeholders drill into any period without touching the underlying gold-layer model.



Customer Insights

48,077 active customers tracked with average spend, order frequency and engagement event metrics.

Behavioral Analytics

Clickstream events (page views, search, add-to-cart, login) reveal how customers move through the funnel.

Geographic & Growth View

Country-level distribution and 2023–2026 customer growth trend support expansion decisions.



Customer sentiment split evenly across positive, neutral and negative, with refund, product, payment and delay issues surfaced by volume.

Agent Performance

Top agents ranked by average resolution time, enabling targeted coaching and workload balancing.

Key Skills

1

Databricks

Lakehouse workspace, Unity Catalog & job orchestration

2

SQL

Complex transformations & gold-layer table creation

3

PySpark

Distributed data processing across bronze → silver

4

Power BI

Executive & operational dashboard design

5

DAX

Calculated measures for KPIs & trend analysis

6

Star Schema

Fact & dimension modeling for analytics

7

ETL

End-to-end extract, transform, load pipelines

8

Data Cleaning

Standardization & quality checks across sources

9

Data Modeling

Relationship design for scalable semantic models

10

Business KPIs

Revenue, customer, support & sentiment metrics